```
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination

# Step 1: Define structure
model = DiscreteBayesianNetwork([
    ('Income_Stability', 'Default_Risk'),
    ('Credit_History', 'Default_Risk'),
    ('Employment_Type', 'Default_Risk')
])

# Step 2: Define CPDs

# Income Stability (0 = Stable, 1 = Unstable)
cpd_income = TabularCPD(
    variable='Income_Stability',
    variable_card=2,
    values=[[0.7], [0.3]]
)

# Credit History (0 = Good, 1 = Bad)
cpd_credit = TabularCPD(
    variable='Credit_History',
    variable_card=2,
    values=[[0.6], [0.4]]
)

# Employment Type (0 = Salaried, 1 = Self-employed)
cpd_employment = TabularCPD(
    variable='Employment_Type',
    variable_card=2,
    values=[[0.65], [0.35]]
)

# Default Risk (0 = Low, 1 = High)
cpd_default = TabularCPD(
    variable='Default_Risk',
    variable_card=2,
    values=[
        # Low Risk
        [0.9, 0.7, 0.6, 0.3, 0.8, 0.5, 0.4, 0.1],
        # High Risk
        [0.1, 0.3, 0.4, 0.7, 0.2, 0.5, 0.6, 0.9]
```

```
    ],
    evidence=['Income_Stability', 'Credit_History', 'Employment_Type'],
    evidence_card=[2, 2, 2]
)

# Step 3: Add CPDs
model.add_cpds(cpd_income, cpd_credit, cpd_employment, cpd_default)

# Step 4: Check model
print("Model Valid:", model.check_model())

# Step 5: Inference
infer = VariableElimination(model)

# Example 1: Full data prediction
result1 = infer.query(
    variables=['Default_Risk'],
    evidence={
        'Income_Stability': 0,
        'Credit_History': 0,
        'Employment_Type': 0
    }
)

print("\nFull Data Prediction:")
print(result1)

# Example 2: Incomplete data (missing Employment Type)
result2 = infer.query(
    variables=['Default_Risk'],
    evidence={
        'Income_Stability': 1,
        'Credit_History': 1
    }
)

print("\nIncomplete Data Prediction:")
print(result2)
```

Case study output

```
Model Valid: True

Full Data Prediction:
+-----------------+---------------------+
| Default_Risk    |   phi(Default_Risk) |
+=================+=====================+
| Default_Risk(0) |              0.9000 |
+-----------------+---------------------+
| Default_Risk(1) |              0.1000 |
+-----------------+---------------------+

Incomplete Data Prediction:
+-----------------+---------------------+
| Default_Risk    |   phi(Default_Risk) |
+=================+=====================+
| Default_Risk(0) |              0.2950 |
+-----------------+---------------------+
| Default_Risk(1) |              0.7050 |
+-----------------+---------------------+
```